

Two-stage attention-guided active learning for nonstationary Gaussian process modeling in robust parameter design

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ABSTRACT

Robust parameter design is vital for continuous quality improvement in industrial practice. However, modeling nonstationary response surfaces with limited data remains challenging because insufficient predictive accuracy may introduce optimistic bias in estimating the optimal parameter settings. We propose a two-stage attention-guided active learning method within a nonstationary Gaussian process (NSGP) modeling framework for robust parameter design. The method employs an acquisition strategy that integrates a softmax-weighted quality loss function with a maximin distance criterion to balance local accuracy around the potential optimum subregion and global space-filling. In the first stage, candidate design points are ranked by their relevance to the potential optimum subregion, and a high-weight subset is selected for further refinement. In the second stage, spatial diversity is enforced by selecting additional points from this subset to mitigate clustering. The selected points are then used to update the training set and the NSGP model. A genetic algorithm is employed to solve the NSGP-based robust parameter design problem and determine the optimal settings. Numerical experiments and engineering case studies show that the proposed method improves predictive accuracy and enhances the robustness of solutions in data-scarce scenarios.

KEYWORDS

Quality engineering; robust parameter design; nonstationary Gaussian process modeling; active learning; two-stage attention-guided acquisition

1. Introduction

1.1. *Research motivation*

In modern manufacturing, product quality is directly linked to enterprise competitiveness, customer loyalty, and operational cost control, and is closely associated with

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improvements in market share and customer satisfaction (Wu and Hamada 2011; He et al. 2021; Li et al. 2025b). Systematic quality improvement further reduces defects, rework, and warranty expenditures (Wu et al. 2020). As a key approach in quality engineering, robust parameter design (RPD) determines the combination of design and process parameters to reduce sensitivity to noise factors, thereby enhancing the consistency and stability of quality characteristics (Bao et al. 2016; Ouyang et al. 2016, 2025). RPD has been widely applied in advanced manufacturing, engineering systems, and service operations, providing theoretical support for performance optimization under uncertainty (Chuang and Wu 2017; An et al. 2020; Huang 2022). Response surface methodology (RSM) is an important tool for implementing RPD by statistically approximating input–output relationships (Fan 2000; Ding et al. 2004; Peterson et al. 2009; Li et al. 2025a). However, the quality characteristics of modern engineering systems increasingly exhibit high dimensionality, strong nonlinearity, and pronounced nonstationarity. Under such conditions, conventional response-surface or surrogate models struggle to achieve adequate accuracy and generalization with limited data. This limitation undermines both the validity of optimal solutions and the efficiency of quality improvement.

Gaussian process (GP/Kriging)–based surrogate modeling has become a cornerstone technique for robust optimization owing to its principled uncertainty quantification and capability for flexible nonlinear function approximation (Williams and Rasmussen 2006; Ankenman et al. 2010; Santner et al. 2018; Liu et al. 2025). Active learning provides an information-gain-oriented design-of-experiments paradigm that enables data-efficient GP modeling and robust optimization under limited data conditions (Wang et al. 2025; Feng et al. 2025). Nevertheless, a large portion of the literature assumes stationarity and overlooks the targeted exploration of subregions near the potential optimum, thereby reducing the efficiency of response surface modeling and compromising the robustness of the obtained solutions. To address this gap, this study develops a two-stage attention-guided active learning framework for NSGP modeling and robust parameter design under limited data. Stage I employs a softmax-weighted quality loss criterion to prioritize predictive accuracy in the potential optimum region; Stage II adopts a maximin space-filling rule to suppress local clustering and improve global coverage. The resulting attention-enhanced NSGP is then embedded within a robust parameter design scheme.

1.2. *Related research work*

In recent years, a substantial body of literature has investigated RPD using GP modeling (Jiang and Tan 2021). Regarding the integration of applications and modeling, Dellino et al. (2012) combined Kriging with nonlinear programming and a resampling strategy to mitigate the uncertainty of surrogate models. Tan and Wu (2012) introduced an expected quadratic loss criterion with a Lugannani–Rice saddlepoint

approximation to enhance optimization reliability. Several studies advanced multiresponse GP modeling within a Bayesian framework (Alshraideh and del Castillo 2014; Kleijnen and Mehdad 2014); Li and Zhou (2016) proposed a pairwise modeling strategy, whereas Ding et al. (2024) developed a covariate-aware Gaussian process profile model, both aiming to improve predictive accuracy. Han and Tan (2016), Wang et al. (2019), and Wang et al. (2022) addressed input uncertainty through closed-form quality loss, IMSE-based sampling, and input location error analysis, respectively. Feng et al. (2021) proposed a Student- t GP for outlier-robust optimization. Chen and Tuo (2022), Barde (2024), and Zhou et al. (2025) addressed high-dimensional problems using projection-pursuit additive Gaussian processes, sparse variational coregionalization models, and Dirichlet Gaussian processes, respectively. Despite these advances, most studies assume fixed training datasets, which restricts the adaptability of design points and undermines the effectiveness of response surface models when only limited samples are available.

Under data-scarce settings, active learning enhances data efficiency and robustness by prioritizing informative samples. Chen et al. (2021) enhanced GP model accuracy by integrating D-optimality with maximin space-filling criteria. Han et al. (2023) developed a dual-acquisition strategy that coordinates the sampling of control and noise variables based on contour estimation. Zhang et al. (2024) employed Gibbs importance sampling in the AK-Gibbs method to better estimate small failure probabilities in high-dimensional problems. However, these approaches often rely on the assumption of stationarity, which limits their ability to capture nonstationary characteristics commonly observed in industrial processes. To address this limitation, a variety of NSGP models have been developed. Gramacy and Lee (2008) introduced treed partition-based NSGP that effectively captured the nonstationary response observed in rocket booster design. Konomi et al. (2014) extended NSGP to multi-output systems for modeling complex fluid dynamics. Rhode (2020) evaluated multiple confidence band construction methods within the NSGP framework. Liu (2021) enhanced the predictive accuracy by integrating global-local GP ensembles with sliding-window ranking mechanisms. With regard to unified modeling and optimization, Davis et al. (2021) proposed the Bayesian composite GP model, which decouples global trends and local fluctuations through non-constant variance structures. Meng et al. (2022) developed the combined framework to balance global exploration and local exploitation in multimodal response surfaces. Deep Gaussian processes (DGPs) have also been adopted for modeling nonstationary functions. Sauer et al. (2022) incorporated DGPs into active learning to approximate complex simulations under data-scarce settings. Gnanasambandam et al. (2024) integrated DGPs with stochastic imputation to improve efficiency across various benchmark functions and metal additive manufacturing applications. However, most existing studies neglect the predictive accuracy of potential optimum regions during active learning, which may lead to overestimation of the robustness of the resulting solutions.

The attention mechanism, originating from deep learning in natural language processing, has been widely adopted in computer vision, time-series forecasting, multi-task optimization, and surrogate modeling for scientific computing (Yu et al. 2022). Its core principle is the adaptive weighting of features to emphasize information most relevant to task performance. In active learning, this principle can guide sampling toward regions exerting strong influence on the response or robustness metrics, thereby improving modeling efficiency and predictive accuracy under limited data. Within GP modeling, attention concepts have been embedded to enhance representation and inference. Li and Ma (2022) established the link between GP mixtures and attention mechanisms, proposing a variant for non-Euclidean spaces. Xie et al. (2022) developed a GP-based channel attention module that probabilistically models channel correlations in an end-to-end manner. Zhong et al. (2024) combined transformers-based learning with Gaussian attention to improve the segmentation of lumber surface defects, particularly for blurred boundaries and irregular patterns. Although these approaches improve model accuracy, they typically adopt a single-objective attention mechanism, which may lead to local clustering of design points and reduce the overall adaptability of the sampling strategy.

To address the above challenges, this study proposes a two-stage attention-guided active learning NSGP modeling framework for robust optimization under limited data. In Stage I, a softmax-weighted quality-loss acquisition focuses sampling on potential optimum subregions to improve local predictive accuracy. In Stage II, a maximin space-filling criterion suppresses local clustering and enhances global coverage. The resulting attention-augmented NSGP surrogate is then embedded in a robust parameter design formulation. The main contributions of this study are as follows:

- (1) Develops a unified attention-guided active-learning NSGP optimization framework with small samples and nonstationary responses, jointly improving nonstationarity characterization and optimization quality.
- (2) Introduces a principled two-stage acquisition policy that couples a target-aware (quality-loss) score with a space-filling (maximin) rule, yielding efficient yet diversity-preserving sampling in the potential optimum regions.
- (3) Designs an objective-driven (quality-loss-based) active-learning procedure that strengthens modeling fidelity and uncertainty control in the potential optimum regions, thereby enabling more accurate and robust solutions.

The remainder of this paper is organized as follows. Section 2 outlines the nonstationary GP modeling approach. Section 3 introduces the attention-augmented robust optimization framework, including the attention weighting scheme, the attention-based active learning loop, and the overall optimization formulation. Section 4 demonstrates the proposed method through two numerical examples and two engineering case studies, with a discussion of its applicability and advantages. Section 5 concludes the paper and outlines future research directions.

2. Nonstationary Gaussian process modeling

Standard GP models typically assume homoscedastic noise and stationary kernels. In many applications, however, responses are nonstationary and heteroscedastic. A heteroscedastic nonstationary GP is therefore considered, in which the length scale $\ell(\mathbf{x})$, signal amplitude $\sigma(\mathbf{x})$, and noise standard deviation $\omega(\mathbf{x})$ are modeled as smooth input-dependent latent functions (Heinonen et al. 2016; Rhode 2020). Based on the Gibbs nonstationary kernel, input dependence is introduced explicitly in the covariance: for any $\mathbf{x}, \mathbf{x}'$, the kernel $k_{\text{NS}}(\mathbf{x}, \mathbf{x}')$ is determined jointly by $\ell(\cdot)$ and $\sigma(\cdot)$; for multivariate inputs, $\ell(\mathbf{x})$ is extended to a diagonal length-scale matrix to accommodate anisotropy. Let $\mathbf{y} = (y_i)_{i=1}^n \in \mathbb{R}^n$ denote observations at inputs $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_n]^\top \in \mathbb{R}^{n \times d}$. The GP regression model is then specified as:

$$y(\mathbf{x}) = f(\mathbf{x}) + \varepsilon(\mathbf{x}), \quad \varepsilon(\mathbf{x}) \sim \mathcal{N}(0, \omega(\mathbf{x})^2), \quad (1)$$

where both the latent function $f(\mathbf{x})$ and the zero-mean observation noise variance $\omega(\mathbf{x})^2$ are unknown and require estimation. A zero-mean GP prior is placed on the latent function $f(\mathbf{x})$:

$$f(\mathbf{x}) \sim \mathcal{GP}(0, K_f(\mathbf{x}, \mathbf{x}')), \quad (2)$$

where $K_f(\mathbf{x}, \mathbf{x}') = \text{cov}(f(\mathbf{x}), f(\mathbf{x}'))$ denotes the covariance between outputs at inputs $\mathbf{x}$ and $\mathbf{x}'$, determined through a kernel function based on input similarity. A nonstationary generalization of the squared exponential kernel is employed (Heinonen et al. 2016):

$$K_f(\mathbf{x}, \mathbf{x}') = \prod_{g=1}^d \sigma(x_g) \sigma(x'_g) \left[\frac{2\ell_g(x_g) \ell_g(x'_g)}{\ell_g^2(x_g) + \ell_g^2(x'_g)} \right]^{1/2} \exp \left(-\frac{(x_g - x'_g)^2}{\ell_g^2(x_g) + \ell_g^2(x'_g)} \right), \quad (3)$$

where $x_g, x'_g \in \mathbb{R}$ are the g -th components of $\mathbf{x}$ and $\mathbf{x}'$, respectively. The functions $\sigma(x_g)$ and $\ell_g(x_g)$ represent the signal standard deviation and the length scale along the g -th dimension. The length-scale, signal variance, and noise variance are modeled as latent functions. Independent GP priors are assigned to the smoothly varying latent functions:

$$\log \ell_g(x_g) \equiv \tilde{\ell}_g(x_g) \sim \mathcal{GP}(\mu_{\ell_g}, K_{\ell_g}(x_g, x'_g)), \quad g = 1, \dots, d, \quad (4)$$

$$\log \sigma_g(x_g) \equiv \tilde{\sigma}_g(x_g) \sim \mathcal{GP}(\mu_{\sigma_g}, K_{\sigma_g}(x_g, x'_g)), \quad g = 1, \dots, d, \quad (5)$$

$$\log \omega(\mathbf{x}) \equiv \tilde{\omega}(\mathbf{x}) \sim \mathcal{GP}(\mu_\omega, K_\omega(\mathbf{x}, \mathbf{x}')). \quad (6)$$

For all three latent processes, the covariance functions take the standard squared exponential form:

$$K_c(x_g, x'_g) = \alpha_c^2 \exp \left(-\frac{(x_g - x'_g)^2}{2\beta_c^2} \right), \quad c \in \{\ell, \sigma, \omega\}. \quad (7)$$

The model is parameterized by nine hyperparameters $\boldsymbol{\theta} = (\mu_\ell, \mu_\sigma, \mu_\omega, \alpha_\ell, \alpha_\sigma, \alpha_\omega, \beta_\ell, \beta_\sigma, \beta_\omega)$, which define the priors for the latent processes $\tilde{\ell}$, $\tilde{\sigma}$, $\tilde{\omega}$. The parameters μ and α typically exert limited influence on model performance, whereas β governs the smoothness of the latent functions and is generally determined from prior knowledge. Given a dataset $(\mathbf{X}, \mathbf{y})$, the model can be equivalently written as $\mathbf{f} \mid \ell, \sigma \sim \mathcal{N}(\mathbf{0}, K_f)$, where $\mathbf{f} = (f(\mathbf{x}_i))_{i=1}^n$ denotes the vector of latent function values at $\mathbf{X}$. $K_f \in \mathbb{R}^{n \times n}$ is computed using Equation (7) with the signal variance vector σ and the length-scale vector ℓ . The observation likelihood is:

$$\mathbf{y} \mid \ell, \sigma \sim \mathcal{N}(\mathbf{0}, K_f + \Omega), \quad (8)$$

where $\Omega = \text{diag}(\omega^2) \in \mathbb{R}^{n \times n}$ is the diagonal noise covariance matrix. $\omega = (\omega(x_i))_{i=1}^n$ denotes the vector of noise standard deviations. Posterior inference is conducted by sampling from the full posterior $p(\mathbf{f}, \tilde{\ell}, \tilde{\sigma}, \tilde{\omega} \mid \mathbf{y}, \theta)$ using Hamiltonian Monte Carlo (HMC) (Gelman et al. 2013; Hoffman and Gelman 2014). Marginalizing $\mathbf{f}$ yields the joint posterior:

$$p(\tilde{\ell}, \tilde{\sigma}, \tilde{\omega} \mid \mathbf{y}) \propto \mathcal{N}(\mathbf{y} \mid \mathbf{0}, K_f + \Omega) \mathcal{N}(\tilde{\ell} \mid \mu_\ell, K_\ell) \mathcal{N}(\tilde{\sigma} \mid \mu_\sigma, K_\sigma) \mathcal{N}(\tilde{\omega} \mid \mu_\omega, K_\omega). \quad (9)$$

The posterior distribution of $\mathbf{f}$ is obtained by integrating out $(\tilde{\ell}, \tilde{\sigma}, \tilde{\omega})$:

$$\begin{aligned} p(\mathbf{f} \mid \mathbf{y}) &= \iiint p(\mathbf{f} \mid \tilde{\ell}, \tilde{\sigma}, \tilde{\omega}, \mathbf{y}) p(\tilde{\ell}, \tilde{\sigma}, \tilde{\omega} \mid \mathbf{y}) d\tilde{\ell} d\tilde{\sigma} d\tilde{\omega} \\ &\approx \frac{1}{m} \sum_{r=1}^m p(\mathbf{f} \mid \tilde{\ell}^{(r)}, \tilde{\sigma}^{(r)}, \tilde{\omega}^{(r)}, \mathbf{y}), \end{aligned} \quad (10)$$

where

$$\tilde{\ell}^{(r)}, \tilde{\sigma}^{(r)}, \tilde{\omega}^{(r)} \sim p(\tilde{\ell}, \tilde{\sigma}, \tilde{\omega} \mid \mathbf{y}) \quad (11)$$

are m posterior HMC samples. For each sample, the conditional posterior of $\mathbf{f}$ is:

$$p(\mathbf{f} \mid \tilde{\ell}^{(r)}, \tilde{\sigma}^{(r)}, \tilde{\omega}^{(r)}, \mathbf{y}) = \mathcal{N}(\boldsymbol{\mu}^{(r)}, \boldsymbol{\Sigma}^{(r)}), \quad (12)$$

The corresponding predictive mean and covariance for the r -th HMC sample are expressed as:

$$\boldsymbol{\mu}^{(r)} = (K_f^{(r)})^\top (K_f^{(r)} + \Omega^{(r)})^{-1} \mathbf{y}, \quad (13)$$

$$\boldsymbol{\Sigma}^{(r)} = K_f^{(r)} - (K_f^{(r)})^\top (K_f^{(r)} + \Omega^{(r)})^{-1} K_f^{(r)}, \quad (14)$$

where $K_f^{(r)}$ denotes the nonstationary kernel matrix computed using $\tilde{\boldsymbol{\ell}}^{(r)}$ and $\tilde{\boldsymbol{\sigma}}^{(r)}$. $\Omega^{(r)}$ represents the diagonal noise covariance matrix derived from $\tilde{\boldsymbol{\omega}}^{(r)}$. Consequently, only the three latent vectors $(\tilde{\boldsymbol{\ell}}, \tilde{\boldsymbol{\sigma}}, \tilde{\boldsymbol{\omega}})$ require HMC sampling, while the posterior of $\mathbf{f}$ can be expressed as a mixture of m Gaussian components. For an arbitrary point, the single-sample predictive distribution of the observation y_* is given by:

$$y_* | \mathbf{y}, \tilde{\boldsymbol{\theta}}^{(r)} \sim \mathcal{N}(\boldsymbol{\mu}_*^{(r)}, \boldsymbol{\Sigma}_*^{(r)} + \boldsymbol{\omega}_*^{(r)2}), \quad \boldsymbol{\omega}_*^{(r)} = \exp(\tilde{\boldsymbol{\omega}}^{(r)}(\mathbf{x}_*)), \quad (15)$$

where $\boldsymbol{\Sigma}_*^{(r)}$ denotes the predictive variance for the latent function and $\boldsymbol{\omega}_*^{(r)}$ corresponds to the predicted noise standard deviation. By aggregating over m HMC samples, the predictive distributions for f_* and y_* can be approximated as:

$$p(f_* | \mathbf{y}) \approx \frac{1}{m} \sum_{r=1}^m \mathcal{N}(\boldsymbol{\mu}_*^{(r)}, \boldsymbol{\Sigma}_*^{(r)}), \quad (16)$$

$$p(y_* | \mathbf{y}) \approx \frac{1}{m} \sum_{r=1}^m \mathcal{N}(\boldsymbol{\mu}_*^{(r)}, \boldsymbol{\Sigma}_*^{(r)} + \boldsymbol{\omega}_*^{(r)2}). \quad (17)$$

Let $\bar{\boldsymbol{\mu}}_*^f = \frac{1}{m} \sum_{r=1}^m \boldsymbol{\mu}_*^{(r)}$, $\bar{\boldsymbol{\Sigma}}_*^f = \frac{1}{m} \sum_{r=1}^m \boldsymbol{\Sigma}_*^{(r)}$, $\bar{\boldsymbol{\omega}}_*^2 = \frac{1}{m} \sum_{r=1}^m \boldsymbol{\omega}_*^{(r)2}$, then the predictive mean and variance of y_* are approximated as:

$$\mathbb{E}[y_* | \mathbf{y}] \approx \bar{\boldsymbol{\mu}}_*^f, \quad \text{Var}(y_* | \mathbf{y}) \approx \bar{\boldsymbol{\Sigma}}_*^f + \bar{\boldsymbol{\omega}}_*^2 + \frac{1}{m} \sum_{r=1}^m (\boldsymbol{\mu}_*^{(r)} - \bar{\boldsymbol{\mu}}_*^f)^2. \quad (18)$$

The predictive mean and variance in Equation (18) are subsequently utilized in constructing the acquisition function and the robust optimization model.

3. Attention-based robust optimization model

3.1. Attention weighting strategy

This section introduces a two-stage attention mechanism for sample selection in active learning. The approach improves the predictive accuracy of Gaussian process regression models near the optimal solution and maintains a reasonable distribution of training samples across the design space. This mechanism draws on the concept of attention, which assigns different weights to inputs in order to identify candidate points with the highest informational value. The attention mechanism operates via a query–key–value framework, whereby candidates exhibiting higher relevance are assigned greater weights. Given a query vector $\mathbf{q} \in \mathbb{R}^{d_k}$, a set of key vectors $\mathbf{K} = [\mathbf{k}_1, \dots, \mathbf{k}_N]^\top \in \mathbb{R}^{N \times d_k}$, and a set of value vectors $\mathbf{V} = [\mathbf{v}_1, \dots, \mathbf{v}_N]^\top \in \mathbb{R}^{N \times d_v}$, the scaled dot-product attention is defined as:

$$\text{Attn}(\mathbf{q}, \mathbf{K}, \mathbf{V}) = \sum_{j=1}^N w_j \mathbf{v}_j, \quad w_j = \text{softmax} \left(\frac{\mathbf{q}^\top \mathbf{k}_j}{\sqrt{d_k} \tau} \right) = \frac{\exp \left(\frac{\mathbf{q}^\top \mathbf{k}_j}{\sqrt{d_k} \tau} \right)}{\sum_{l=1}^N \exp \left(\frac{\mathbf{q}^\top \mathbf{k}_l}{\sqrt{d_k} \tau} \right)}, \quad (19)$$

where $\tau > 0$ denotes the temperature parameter ($\tau \downarrow$ results in a sharper distribution), and $\sqrt{d_k}$ is a scaling factor introduced to mitigate numerical amplification in high-dimensional dot products. For improved numerical stability, the softmax in Equation (19) is commonly written as:

$$w_j = \frac{\exp \left(\frac{\mathbf{q}^\top \mathbf{k}_j}{\sqrt{d_k} \tau} - m \right)}{\sum_{l=1}^N \exp \left(\frac{\mathbf{q}^\top \mathbf{k}_l}{\sqrt{d_k} \tau} - m \right)}, \quad m = \max_l \frac{\mathbf{q}^\top \mathbf{k}_l}{\sqrt{d_k} \tau}, \quad (20)$$

where $\sum_{j=1}^N w_j = 1$ and $w_j \geq 0$. Within active learning, the attention mechanism yields a unified “scoring (relevance)–normalization (softmax)–selection” paradigm. Different relevance objectives (e.g., focusing on the potential optimum region or achieving space-filling) can be handled by designing $\mathbf{q}$ and $\mathbf{k}_j$ accordingly, followed by softmax to obtain w_j for ranking and selection. This work employs a two-stage strategy: Stage I focuses on the potential subregion containing the optimal solution, and Stage II enhances spatial coverage.

3.2. Active learning method based on the attention mechanism

The proposed attention-based active learning algorithm consists of the following main steps: (1) Initial model training: Construct the NSGP model structure, generate the initial training samples, and fit the initial NSGP model. (2) Candidate point selection: Employ the quality-loss and maximin space-filling criteria to construct a two-

stage attention-based acquisition function. (3) Acquisition score computation: First, compute the quality-loss weights and select high-weight points to be added to the candidate set; then, compute the distance weights and select the highest-weight points to augment the training set. (4) Model update: Update the NSGP model using the updated training set and return to Step (2) until the stopping criterion is satisfied. Specifically, let the candidate pool be $\mathcal{X}_{\text{pool}} = \{\mathbf{x}_j\}_{j=1}^N$, and the current training set be $\mathbf{S}_t = \{\mathbf{x}_i\}_{i=1}^{n_t}$. Based on the NSGP model, the predictive mean and variance at any point $\mathbf{x}$ are denoted by $\mu(\mathbf{x})$ and $v(\mathbf{x})$, respectively. The target response is T . The two-stage attention mechanism is then defined as follows:

In the first stage, to rapidly identify the potential subregion containing the optimal solution, the acquisition function is constructed using the quality loss criterion, thereby prioritizing this region (where $\mu(\mathbf{x})$ is close to T and the uncertainty is small). The “quality feature” is first constructed as the keys:

$$QL_j = (\mu(\mathbf{x}_j) - T)^2 + v(\mathbf{x}_j). \quad (21)$$

Since the metric in Equation (21) is smaller for better candidates, whereas the softmax in Equation (20) typically assumes larger values are preferred, the quality loss function is negated: $s_j^{(1)} = -QL_j$. The scaled dot-product form of the quality loss is expressed as:

$$s_j^{(1)} = \frac{\mathbf{q}^{(1)\top} \mathbf{k}_j^{(1)}}{\sqrt{d_k^{(1)}}}, \quad \mathbf{k}_j^{(1)} = W_k^{(1)} \left[-(\mu(\mathbf{x}_j) - T)^2, -\lambda v(\mathbf{x}_j) \right]^\top, \quad \mathbf{q}^{(1)} = W_q^{(1)} [1, 1]^\top, \quad (22)$$

where $\lambda \geq 0$ is a trade-off coefficient, set to 1 in this study, and $d_k^{(1)} = \dim(\mathbf{k}_j^{(1)}) = \dim(\mathbf{q}^{(1)})$. Applying the softmax to Equation (22) yields the first-stage attention weights:

$$\alpha_j = \text{soft max} \left(\frac{s_j^{(1)}}{\tau_1} \right) = \frac{\exp(-QL_j/\tau_1)}{\sum_{l=1}^N \exp(-QL_l/\tau_1)}, \quad j = 1, \dots, N, \quad (23)$$

where $\tau_1 > 0$ controls the sharpness of the distribution. The top N_1 candidates with the highest weights are selected to form the first-stage candidate set:

$$\mathbf{X}_{\text{ready}} = \text{TopK}(\mathcal{X}_{\text{pool}}, \alpha, N_1), \quad (24)$$

where $\mathcal{X}_{\text{pool}}$ is the candidate pool. $\mathbf{X}_{\text{ready}}$ denotes the candidate set produced in the first stage. N_1 controls the attention breadth: smaller values intensify the emphasis on quality loss, whereas larger values broaden exploration. N_1 is set to 10 to place greater focus on the potential optimum subregion.

The second stage then computes the attention weights for space-filling. The goal is to select the point with the best space-filling performance from $\mathbf{X}_{\text{ready}}$ to avoid local clustering and stabilize covariance estimation. The minimum Euclidean distance between a design point $\mathbf{x}_j$ and the current training set is defined as:

$$d_{\min}(\mathbf{x}_j, \mathbf{S}_t) = \min_{\mathbf{x}_i \in \mathbf{S}_t} \|\mathbf{x}_j - \mathbf{x}_i\|_2. \quad (25)$$

Since the distance in Equation (25) is larger for better candidates, it can be directly expressed in scaled dot-product form:

$$s_j^{(2)} = \frac{\mathbf{q}^{(2)\top} \mathbf{k}_j^{(2)}}{\sqrt{d_k^{(2)}}}, \quad \mathbf{k}_j^{(2)} = W_k^{(2)}[d_{\min}(\mathbf{x}_j, \mathbf{S}_t)], \quad \mathbf{q}^{(2)} = W_q^{(2)}[1], \quad (26)$$

where $d_k^{(2)} = \dim(\mathbf{k}_j^{(2)}) = \dim(\mathbf{q}^{(2)})$. The score $s^{(2)}$ in Equation (26) is then normalized using the softmax:

$$\beta_j = \frac{\exp(s_j^{(2)}/\tau_2)}{\sum_{\mathbf{x}_\ell \in \mathbf{X}_{\text{ready}}} \exp(s_\ell^{(2)}/\tau_2)}, \quad \mathbf{x}_j \in \mathbf{X}_{\text{ready}}, \quad (27)$$

where the temperature parameter $\tau_2 > 0$ controls the preference for points farthest from the existing samples: smaller τ_2 leads to weights more concentrated on the most distant candidate. Without loss of generality, both temperatures are set to $\tau_1 = \tau_2 = 0.5$. The top N_2 points ranked by β_j are selected from $\mathbf{X}_{\text{ready}}$:

$$\mathbf{X}_{\text{new}} = \text{TopK}(\mathbf{X}_{\text{ready}}, \beta, N_2), \quad (28)$$

where $\mathbf{X}_{\text{new}}$ denotes the second-stage selected design points, which are subsequently used to update the training set. The new experimental points obtained from Equation (28) lie within the potential optimum subregion while maintaining good space-filling properties in this region, thereby substantially improving sample utilization efficiency. The overall attention-based active learning NSGP modeling process is summarized in Figure 1 and Algorithm 1.

3.3. Proposed robust optimization method

The procedure of the proposed two-stage attention-based active learning method for NSGP modeling and parameter optimization is illustrated in Figure 2.

Step 1: Fit the initial NSGP model using the initial training samples.

Step 2: Construct the quality loss function and the spatial distance function using the outputs of the initial model from Equations (17)–(18).

Algorithm 1 Attention-based active learning NSGP modeling procedure.

Input: Maximum iterations $N_{\max}$; points per iteration N_2 ; initial training set $\mathbf{S}_t$; candidate pool $\mathcal{X}_{\text{pool}}$.

Output: Final training set and updated NSGP model.

- 1: Fit the initial NSGP model using $\mathbf{S}_t$.
 - 2: **for** $s \leftarrow 1$ to $N_{\max}$ **do**
 - 3: Fit (or update) the NSGP model with the current $\mathbf{S}_t$.
 - 4: Compute predictive mean $\mu(\mathbf{x})$ and variance $v(\mathbf{x})$ for all $\mathbf{x} \in \mathcal{X}_{\text{pool}}$.
 - 5: Compute quality-loss weights $\{\alpha_j\}$ on $\mathcal{X}_{\text{pool}}$ using Eq. (23), and form $\mathbf{X}_{\text{ready}}$ (e.g., by TOPK with N_1).
 - 6: **for** $i \leftarrow 1$ to N_2 **do**
 - 7: Compute space-filling weights $\{\beta_j\}$ on $\mathbf{X}_{\text{ready}}$ using Eq. (27).
 - 8: Select $\mathbf{x}_* \leftarrow \arg \max_{\mathbf{x}_j \in \mathbf{X}_{\text{ready}}} \beta_j$.
 - 9: $\mathbf{S}_t \leftarrow \mathbf{S}_t \cup \{\mathbf{x}_*\}$; $\mathcal{X}_{\text{pool}} \leftarrow \mathcal{X}_{\text{pool}} \setminus \{\mathbf{x}_*\}$.
 - 10: Update the NSGP model with the new $\mathbf{S}_t$.
 - 11: **end for**
 - 12: **end for**
-

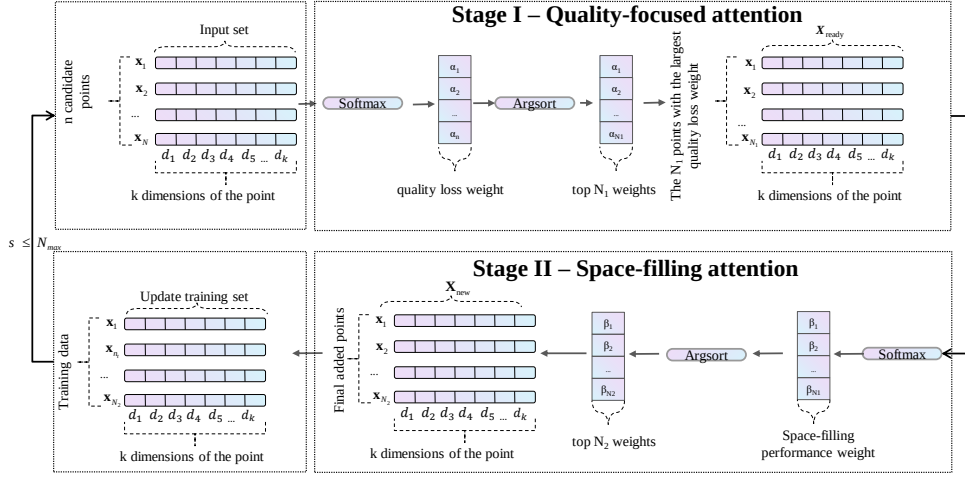


Figure 1. Attention-based active learning NSGP modeling procedure.

- Step 3:** Combine the two metrics from Step 2 to construct the two-stage attention-based acquisition function.
- Step 4:** Apply the active learning process described in Algorithm 1 to select new design points.
- Step 5:** Update the training dataset and the NSGP model, and obtain the predicted response mean and variance.
- Step 6:** Formulate the quality loss function as the robust optimization objective using the predicted mean and variance from Step 5.
- Step 7:** Employ the genetic algorithm (GA) to search for the global optimum within the feasible domain of the input variables, thereby determining the optimal parameter settings.

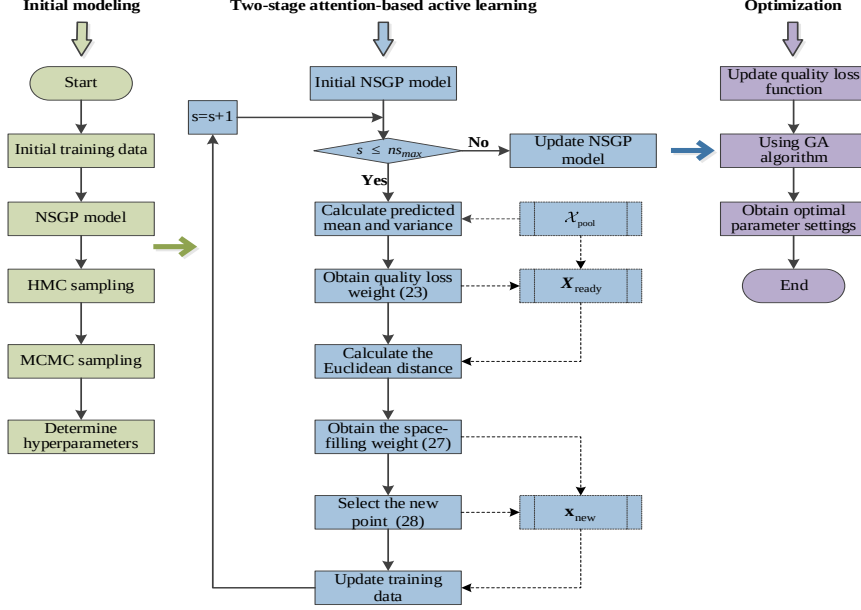


Figure 2. Flowchart of the proposed NSGP modeling and parameter optimization method.

4. Case study

4.1. Two-dimensional numerical example

This section validates the effectiveness of the proposed method using a two-dimensional test function. The function is a piecewise-defined response surface exhibiting pronounced nonstationarity in the two-dimensional input space. The mathematical expression of $f(x_1, x_2)$ is given by:

$$f(x_1, x_2) = \begin{cases} (2 + 0.5x_2) \sin(x_1) + x_2^2, & \text{if } x_1 > 0, \\ x_1^2 + x_2^2 + (8 - (x_1^2 + x_2^2)) (1 - e^{-x_1^2}), & \text{if } x_1 \leq 0, \end{cases} \quad (29)$$

where $(x_1, x_2) \in [-2, 2]^2$. For $x_1 > 0$, the function exhibits periodic oscillations along the x_1 direction, whereas for $x_1 \leq 0$, it presents a circularly symmetric and smooth response. This behavior reflects significant nonstationarity and structural discontinuity. To further assess the local predictive performance of the model in the target region, a specific target value f^* is defined for this test function. Around this target value, a region of interest $\mathcal{R}^t$ is defined as the set of input points satisfying:

$$\mathcal{R}^t = \{(x_1, x_2) \in \mathcal{X} \mid |f(x_1, x_2) - f^*| < t\}, \quad (30)$$

where $t > 0$ is a prescribed threshold. The set $\mathcal{R}^t$ can be regarded as the potential optimum subregion of the design space, within which the predictive accuracy is of critical importance for robust optimization. In this study, the target value is set to

$f^* = 1.5$, which is deliberately chosen to avoid the extrema and thereby prevent misleading effects caused by extreme-value behavior. To quantify the discrepancy between predictions and the true values, the root mean square error (RMSE) is adopted as the performance metric, defined as:

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n \left(\hat{f}(x_1^i, x_2^i) - f(x_1^i, x_2^i) \right)^2}, \quad (31)$$

where $\hat{f}(x_1^i, x_2^i)$ denotes the model prediction, $f(x_1^i, x_2^i)$ is the true value, and n is the total number of test samples. In this experiment, 20 initial design samples are first generated using Latin hypercube sampling (LHS) and employed to fit the initial NSGP model. Subsequently, 20 additional samples are selected using the proposed two-stage attention-based active learning method, resulting in a final set of 40 design points. The NSGP model is then updated using this final training data. For a direct comparison, several widely used experimental design methods are adopted as baseline approaches: the G_optimality (G_opt) method (Omwando 2024), the D_optimality (D_opt) method (Lichota 2016), the LHS method (Guo et al. 2023), and the centroidal Voronoi tessellations (CVT) method (Vassiliades et al. 2018). All methods use the same total number of samples. The spatial distributions of the 40 design points obtained by the different methods in the two-dimensional design space are illustrated in Figure 3.

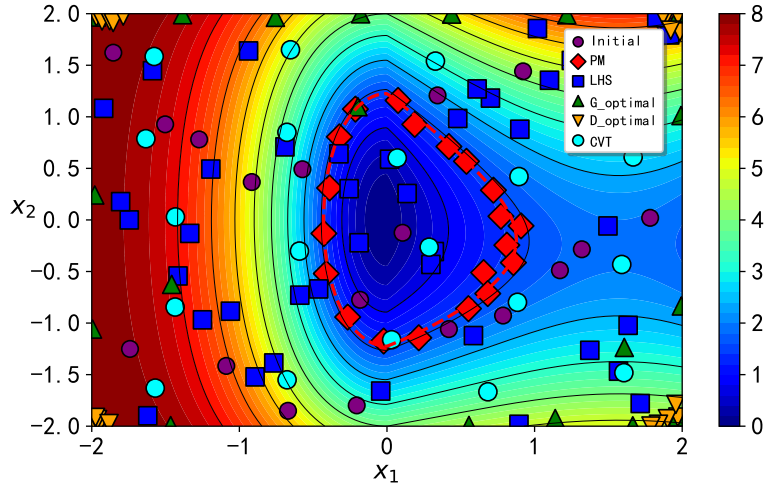


Figure 3. Spatial distribution of the 40 design points obtained by different methods.

In Figure 3, the purple circles represent the initial design points, whereas the red diamonds indicate the points selected using the proposed active learning method. The red dashed line depicts the contour corresponding to the target value of 1.5. As shown, the design points selected by the proposed method (PM) are predominantly concentrated near the target-value contour. This distribution suggests that the algorithm tends to allocate experimental points to the potential optimum region. Furthermore, the design

points close to the target contour exhibit a more uniform spatial distribution, with no apparent overlap or clustering. This characteristic demonstrates the favorable space-filling capability of the proposed active learning method. To further demonstrate the predictive accuracy advantage of the proposed approach, the RMSE values within the potential optimum region are evaluated. Comparisons are made across different total training sample sizes (40–100) and different numbers of newly added design points (20 and 10). For each method, the RMSE computation is repeated 30 times to ensure the reliability of the comparison. The results are summarized in Tables 1–2 and Figure 4. Specifically, Tables 1 and 2 report both the RMSE values and the corresponding improvement ratios of the proposed method over alternative approaches, based on 10 and 20 active learning design points, respectively. Other comparative results, including the maximum, minimum, first quartile (Q1), and third quartile (Q3), are provided in Appendix A.

Table 1. RMSE results and PM improvement ratio with 10 active learning design points (PM improvement for PM is undefined and shown as “—”).

Number	RMSE (Mean)					RMSE (Median)				
	D_opt	G_opt	LHS	CVT	PM	D_opt	G_opt	LHS	CVT	PM
40	0.0919	0.0894	0.0780	0.0651	0.0562	0.0854	0.0823	0.0675	0.0639	0.0503
50	0.0621	0.0650	0.0565	0.0560	0.0386	0.0621	0.0663	0.0539	0.0586	0.0382
60	0.0514	0.0540	0.0451	0.0532	0.0369	0.0487	0.0548	0.0448	0.0430	0.0362
70	0.0427	0.0442	0.0392	0.0365	0.0317	0.0430	0.0419	0.0349	0.0357	0.0325
80	0.0369	0.0416	0.0330	0.0333	0.0246	0.0365	0.0390	0.0320	0.0309	0.0241
90	0.0308	0.0323	0.0261	0.0296	0.0216	0.0288	0.0317	0.0266	0.0264	0.0204
100	0.0225	0.0249	0.0238	0.0252	0.0181	0.0228	0.0250	0.0228	0.0232	0.0192

Number	PM Improvement (Mean, %)					PM Improvement (Median, %)				
	D_opt	G_opt	LHS	CVT	PM	D_opt	G_opt	LHS	CVT	PM
40	38.85%	37.14%	27.95%	13.67%	—	41.10%	38.88%	25.48%	21.28%	—
50	37.84%	40.62%	31.68%	31.07%	—	38.49%	42.38%	29.13%	34.81%	—
60	28.21%	31.67%	18.18%	30.64%	—	25.67%	33.94%	19.20%	15.81%	—
70	25.76%	28.28%	19.13%	13.15%	—	24.42%	22.43%	6.88%	8.96%	—
80	33.33%	40.87%	25.45%	26.13%	—	33.97%	38.21%	24.69%	22.01%	—
90	29.87%	33.13%	17.24%	27.03%	—	29.17%	35.65%	23.31%	22.73%	—
100	19.56%	27.31%	23.95%	28.17%	—	15.79%	23.20%	15.79%	17.24%	—

Based on the results in Table 1 and Figures 4(a-b), when ten active learning design points are used, the PM method achieves the lowest mean RMSE in the potential optimum region across all training sample sizes. When the total number of experimental samples is 40, the PM method attains a mean RMSE of 0.0562. This is approximately 13.67% lower than the CVT method (0.0651) and about 38.85% lower than the D_opt method (0.0919). The advantage remains as the sample size increases. When the total number of samples reaches 100, the mean RMSE of the PM method further decreases to 0.0181, representing a 19.56% reduction compared with the D_opt method (0.0225) and a 28.17% reduction compared with the CVT method (0.0252). Among all comparisons, the largest improvement is observed when the training sample size is 80,

Table 2. RMSE results and PM improvement ratio with 20 active learning design points (PM improvement for PM is undefined and shown as “—”).

Number	RMSE (Mean)					RMSE (Median)				
	D_opt	G_opt	LHS	CVT	PM	D_opt	G_opt	LHS	CVT	PM
40	0.1517	0.0954	0.0730	0.0817	0.0540	0.1385	0.0890	0.0644	0.0802	0.0310
50	0.0990	0.0530	0.0538	0.0719	0.0324	0.0954	0.0530	0.0510	0.0730	0.0249
60	0.0499	0.0481	0.0396	0.0565	0.0216	0.0481	0.0458	0.0390	0.0562	0.0207
70	0.0453	0.0410	0.0359	0.0511	0.0235	0.0466	0.0422	0.0356	0.0503	0.0222
80	0.0345	0.0400	0.0359	0.0419	0.0223	0.0367	0.0378	0.0347	0.0431	0.0205
90	0.0319	0.0298	0.0263	0.0391	0.0187	0.0298	0.0310	0.0263	0.0377	0.0180
100	0.0281	0.0321	0.0227	0.0354	0.0173	0.0281	0.0328	0.0230	0.0352	0.0164

Number	PM Improvement (Mean, %)					PM Improvement (Median, %)				
	D_opt	G_opt	LHS	CVT	PM	D_opt	G_opt	LHS	CVT	PM
40	64.40%	43.40%	26.03%	33.90%	—	77.62%	65.17%	51.86%	61.35%	—
50	67.27%	38.87%	39.78%	54.94%	—	73.90%	53.02%	51.18%	65.89%	—
60	56.71%	55.09%	45.45%	61.77%	—	56.96%	54.80%	46.92%	63.17%	—
70	48.12%	42.68%	34.54%	54.01%	—	52.36%	47.39%	37.64%	55.86%	—
80	35.36%	44.25%	37.88%	46.78%	—	44.14%	45.77%	40.92%	52.44%	—
90	41.38%	37.25%	28.90%	52.17%	—	39.60%	41.94%	31.56%	52.25%	—
100	38.43%	46.11%	23.79%	51.13%	—	41.64%	50.00%	28.70%	53.41%	—

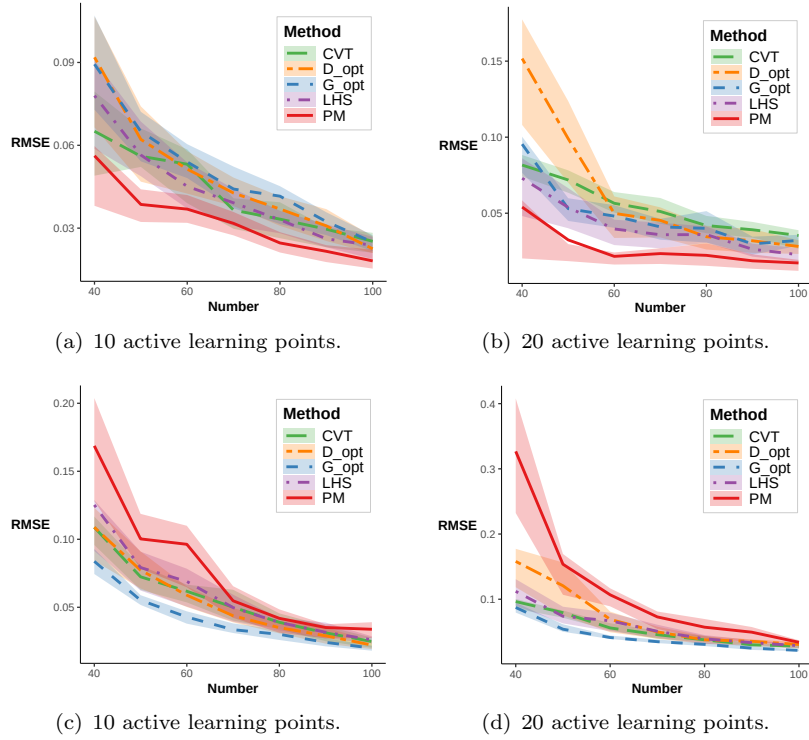


Figure 4. Comparison of RMSE with different numbers of active learning design points: (a)–(b) potential optimum subregion; (c)–(d) overall response surface.

compared with the D_opt method, with a gain of 40.87%. The smallest improvement occurs when the training sample size is 70, compared with the CVT method, with a gain of 13.15%. The median RMSE reflects the typical performance of the model. The

PM method also performs strongly under this metric, indicating the high robustness of its results. With ten active learning design points and a total of 40 experimental samples, the median RMSE of the PM method is 0.0503. This is approximately 21.28% lower than that of the CVT method (0.0639) and about 41.10% lower than that of the D_opt method (0.0854). When the total number of experimental samples reaches 100, the PM method achieves a median RMSE of 0.0192. This represents a 15.79% reduction compared with both the D_opt and LHS methods (0.0228 each), and a 23.20% reduction compared with the G_opt method (0.0250). The largest improvement is observed when the training sample size is 50 compared with the G_opt method, with a gain of 42.38%. The smallest improvement occurs when the training sample size is 70 compared with the LHS method, with a gain of 6.88%. From Table 2 and Figures 4(a-b), when 20 active learning samples are added, the PM method outperforms the alternatives in terms of RMSE across all cases. The advantage is particularly evident in the early learning stage when the total sample size is small. With a total of 40 experimental samples, the PM method achieves a mean RMSE of 0.0540, which is approximately 26.03% lower than LHS (0.0730) and 64.40% lower than D_opt (0.1517). When the total sample size increases to 100, the mean RMSE of the PM method decreases to 0.0173, which remains the lowest in Table 2. For this sample size, the value is 23.79% lower than LHS (0.0227) and 51.13% lower than CVT (0.0354). Across all sample sizes, the largest improvement is observed when the sample size is 50 compared with D_opt, with a gain of 67.27%. The smallest improvement is observed when the sample size is 100 compared with LHS, with a gain of 23.79%.

When the number of active learning samples is 20, the PM method also demonstrates a clear advantage in terms of the median RMSE. With a total sample size of 40, the median RMSE is 0.0310. This is 51.86% lower than the second-best LHS method (0.0644) and 77.62% lower than the D_opt method (0.1385). When the total sample size increases to 100, the median RMSE of the PM method is 0.0164, the lowest among all methods. At this size, it is 28.70% lower than the LHS method (0.0230) and 53.41% lower than the CVT method (0.0352). The largest improvement occurs with 40 training samples compared with the D_opt method, reaching 77.62%. The smallest improvement is seen with 100 training samples compared with the LHS method, at 28.70%.

The comparative results in the tables and figures indicate that the proposed PM method achieves a clear advantage in predictive accuracy within the potential optimum region. This advantage mainly arises from its two-stage attention mechanism. In the first stage, design points are selected to substantially improve predictive accuracy in the potential optimum region. In the second stage, a spatial-distance criterion is applied to suppress local clustering, thereby enhancing space-filling in that region. By contrast, the G_opt method improves prediction by minimizing the maximum prediction variance across the design space. The D_opt method seeks to optimize overall parameter estimation accuracy by maximizing the determinant of the information

matrix. The LHS method focuses on evenly distributing samples in each input dimension to achieve full coverage of the design space. The CVT method ensures uniform sampling density by moving sample points toward the centroids of their Voronoi cells. Consequently, under various conditions, these methods yield lower predictive accuracy in the potential optimum region than the PM method.

It is noteworthy that, in terms of the overall response surface prediction accuracy (Figures 4(c-d)), the RMSE of the PM method is slightly higher than that of the alternative approaches. This difference occurs because the PM method allocates new design points to the most promising potential optimum region for robust optimization. Such allocation inevitably sacrifices part of the global prediction accuracy. However, the results in Figure 4 show that the decrease in global accuracy is much smaller than the improvement achieved in the potential optimum region. Robust optimization places greater emphasis on prediction accuracy within the optimum region. Therefore, this local improvement is more important for obtaining high-quality and stable solutions. Although the PM method incurs a small loss in global accuracy, it significantly enhances the prediction accuracy in the potential optimum region. This improvement enhances the quality, stability, and validity of solutions in robust parameter design.

4.2. *Five-dimensional numerical example*

In practical applications, multiple variables often exhibit complex and nonlinear interactions. To evaluate the applicability and advantages of the proposed method in high-dimensional settings, a five-dimensional piecewise-defined numerical test function is considered:

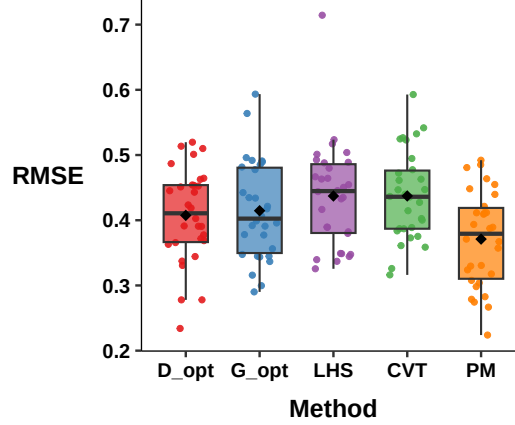
$$y = \begin{cases} 2(\sin(\pi x_1) + \cos(2\pi x_2) + x_3^2 + e^{-x_4} + x_5 + x_3 - 1), & \text{if } x_1 + x_2 > 1, \\ 2(\sin(\pi x_1) + \cos(2\pi x_2) + x_3^2 + e^{-x_4} + x_5 - 1), & \text{if } x_1 + x_2 \leq 1, \end{cases} \quad (32)$$

where $x_g \in [0, 1]$, $g = 1, 2, \dots, 5$. The function exhibits a structural change between the two subregions $x_1 + x_2 > 1$ and $x_1 + x_2 \leq 1$, leading to distinct output behaviors and representing typical input-dependent nonstationarity. In this example, the target value is set to 5.5, and the optimum region is defined as $5 \leq y \leq 6$. The RMSE is employed to evaluate predictive accuracy within this region. The experimental procedure is as follows: first, 80 initial experimental samples are selected using the LHS method to fit the NSGP model; second, 20 additional samples are generated using the proposed two-stage active learning method with the attention mechanism, yielding a total of 100 samples; finally, this process is repeated 30 times, and the RMSE results in the optimum region are compared with those of multiple alternative methods. The results are reported in Table 3 and Figure 5.

As shown in Table 3, the PM method attains a mean RMSE of 0.3711, outperform-

Table 3. RMSE results within the optimum region ($5 \leq y \leq 6$)

Method	Mean	Median	Min	Max	Q1	Q3
D_opt	0.4075	0.4107	0.2340	0.5197	0.3665	0.4539
G_opt	0.4146	0.4025	0.2901	0.5934	0.3497	0.4806
CVT	0.4372	0.4360	0.3162	0.5928	0.3870	0.4762
LHS	0.4372	0.4447	0.3256	0.7144	0.3804	0.4860
PM	0.3711	0.3793	0.2239	0.4921	0.3102	0.4189

**Figure 5.** Comparison of RMSE results within the optimum region ($5 \leq y \leq 6$).

ing all alternative approaches. The D_opt method ranks second with 0.4075, whereas the G_opt method records the highest mean RMSE at 0.4146. Relative to D_opt and G_opt, the PM method achieves reductions of approximately 8.93% and 10.49%, respectively. Compared with LHS and CVT, both with a mean RMSE of 0.4372, the reduction reaches 15.12%. For the median RMSE, the PM method again delivers the best performance (0.3793), representing reductions of 5.76% compared with G_opt (0.4025), 7.65% compared with D_opt (0.4107), 13.00% compared with LHS (0.4360), and 14.71% compared with CVT (0.4447). The consistent advantage in both mean and median RMSE underscores the stability and overall superiority of the PM method. This performance is driven by its two-stage active learning framework with an attention mechanism. In the first stage, a quality loss-based acquisition function is applied to refine the response surface representation within the optimum region. In the second stage, the design maximizes the minimum spatial distance, reducing local clustering and improving space-filling in the region. This design substantially increases the efficiency of experimental resource utilization in robust optimization and yields a more effective nonstationary response surface model.

4.3. Lithium battery state-of-charge prediction

This section adopts the lithium battery state-of-charge (SoC) prediction case study presented by Mustafa et al. (2024) to evaluate the performance of the proposed method. During different stages of battery discharge, the SoC exhibits markedly different response rates to current and voltage, demonstrating typical nonstationary behavior. Using an 18650-type lithium-ion battery as the study object, the case involves one output variable—SoC (%). SoC is defined as the percentage of the current charge relative to the total capacity of the battery, ranging from $y \in [0, 100]$, and serves as a key indicator for evaluating the remaining capacity of the battery. This case study inputs include three variables: temperature x_1 ($^{\circ}\text{C}$), current x_2 (A), and voltage x_3 (V). In this experiment, the SoC target value is set to 30, and the optimum region is defined as $25 \leq y \leq 35$. First, 180 initial training points are generated using the LHS method. The additional 30 points are then selected with the proposed two-stage active learning algorithm, resulting in a total of 210 experimental samples. The entire process is repeated 20 times to ensure robustness of the results. Table 4 and Figure 6 show the RMSE results for the optimum region from different methods.

Table 4. RMSE results from 20 runs.

Method	Mean	Median	Max	Min	Q1	Q3
G_opt	7.9793	6.8957	18.3144	5.5627	6.0980	8.3195
D_opt	7.8077	7.0209	15.5626	5.9420	6.7881	7.6917
CVT	10.7557	9.2375	26.6414	6.9491	7.2289	12.0859
LHS	9.2725	7.6885	16.9593	6.4247	7.0761	10.5335
PM	6.9880	6.7256	10.9925	4.8605	6.1098	7.3933

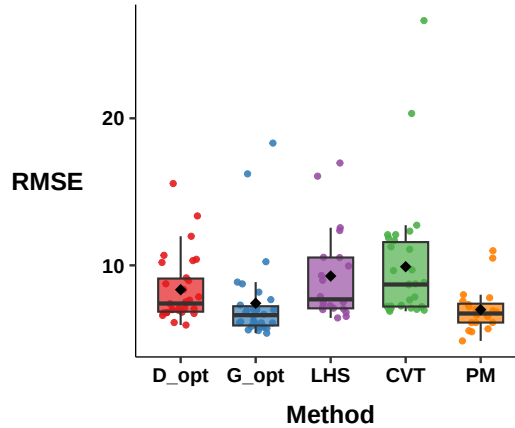


Figure 6. Comparison of battery SoC prediction results.

Table 4 and Figure 6 show that the PM method achieves the highest predictive accuracy within the optimum region. Its mean RMSE is 6.9880, which is markedly lower than all baseline methods. The CVT method records the highest mean RMSE

(10.7557), with a gap of 35.03% compared with the PM method, while the second-best D_opt method (7.8077) still lags by 10.50%. The median RMSE results align with the mean trends: the PM method again attains the lowest value (6.7256), 27.19% lower than that of CVT (9.2375), indicating its ability to maintain high predictive accuracy in most trials. Beyond central tendency, the PM method also excels in controlling extreme values. Its maximum RMSE is only 10.9925, which is 58.74% lower than the CVT method's 26.6414, demonstrating effective suppression of extreme errors. Its minimum RMSE of 4.8605 outperforms all other methods, being 30.05% lower than the minimum RMSE of CVT (6.9491), suggesting a greater tendency to converge to high-accuracy solutions during the search process. In terms of stability, the PM method achieves an interquartile range (IQR) of 1.2835, only 26.43% of that of CVT (4.8570), representing a 73.57% reduction. This further confirms that the PM method attains high robustness without compromising accuracy. As illustrated in Figure 6, the RMSE distribution of the PM method is more concentrated, with fewer outliers, validating its combined advantage in predictive accuracy and robustness within the optimum region.

4.4. Robust parameter design for a micro-rocket

This section adopts the micro-rocket design case study introduced by Box, et al. (2011) to validate the effectiveness and advantages of the proposed method in product quality design. The rocket model consists of a nose cone, fuselage, launch damper lock with parachute, inner tube, and stabilizing fins, as shown in Figure 7. The output response is the maximum flight height (MH) y (m), which serves as a key indicator for evaluating the overall performance of the rocket. Analyzing the maximum altitude provides an integrated measure of thrust, aerodynamic drag, and structural design performance. In this case, the target value is set to 400 m, representing a typical nominal-the-best quality characteristic. The simulated flight trajectory and the relationship between maximum flight altitude and nose cone diameter are presented in Figures 8(a) and 8(b), respectively.

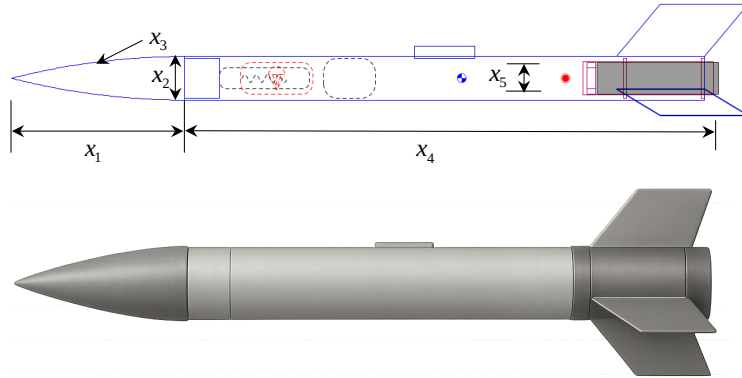


Figure 7. Rocket model diagram.

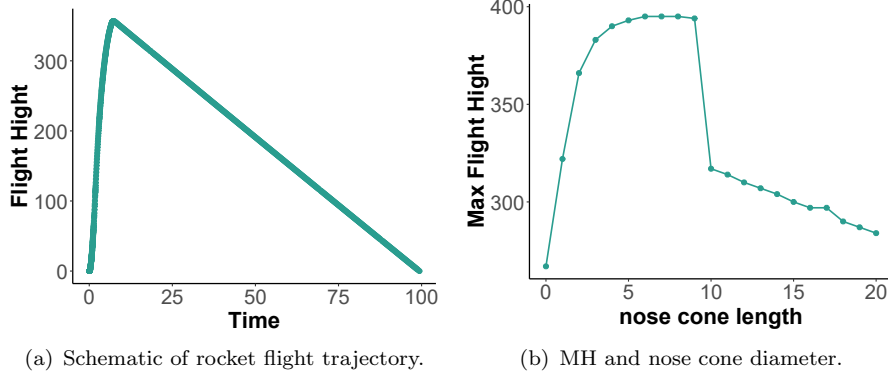


Figure 8. Simulated flight trajectory and the relationship between MH and nose cone diameter.

The case study involves five continuous input parameters: nose cone length x_1 (cm), base diameter of the nose cone x_2 (cm), nose cone wall thickness x_3 (cm), fuselage length x_4 (cm), and inner diameter of the fuselage x_5 (cm). Their respective ranges are $x_1 \in [0, 20]$, $x_2 \in [2, 5]$, $x_3 \in [0, 1]$, $x_4 \in [30, 50]$, and $x_5 \in [1, 2]$. In addition, there is one binary discrete variable—the nose cone shape x_6 —where $x_6 = 1$ denotes an ogive with a cut tip, and $x_6 = 0$ denotes an ogive with a truncated tip. According to the design specifications, for longer nose cones ($x_1 > 10$ cm), the cut-tip ogive is preferred to reduce shock wave drag during supersonic flight, thereby improving aerodynamic performance and enhancing stability. For shorter nose cones ($x_1 < 10$ cm), the truncated-tip ogive is selected to increase lift and improve stability in subsonic flight. This shape transition at $x_1 = 10$ cm introduces a distinct structural change, causing a step-like behavior in the output response at this location and resulting in a pronounced nonstationary characteristic.

The objective of robust parameter design in this case study is to identify an optimal combination of design parameters that enables the rocket to achieve a maximum flight altitude of 400 m or as close as possible to this target. To this end, the subsequent experiments employ the proposed method to improve the predictive accuracy within the potential optimum region, thereby constructing a more efficient and stable nonstationary response surface model. The experimental procedure is as follows: first, 30 initial training points are selected using the LHS method, and an initial NSGP-based nonstationary response surface model is fitted. Next, 20 additional design points are generated using the proposed two-stage active learning algorithm, bringing the total number of training samples to 50, as shown in Table 5.

As shown in Table 5, the proposed method maintains the response values of the newly added 31st–50th experimental points close to the target of 400. Moreover, these points are evenly distributed within the design space, without noticeable clustering or overlapping. This outcome verifies the effectiveness of the proposed two-stage active learning modeling approach in achieving both space-filling and target-focused sampling. Based on the experimental data in Table 5, an NSGP model is constructed, and

Table 5. Experimental results of the PM method.

Runs 1–25								Runs 26–50							
Run	x_1	x_2	x_3	x_4	x_5	x_6	y	Run	x_1	x_2	x_3	x_4	x_5	x_6	y
1	15.5	3.13	0.96	48.8	1.89	1	26.7	26	9.9	2.20	0.02	38.4	1.62	0	262
2	12.9	2.81	0.07	41.3	1.96	1	130	27	14.2	2.91	0.28	38.6	1.86	1	87.5
3	1.3	2.06	0.63	37.4	1.31	0	227	28	16.6	2.71	0.90	40.9	1.38	1	50.5
4	0.5	3.97	0.10	45.6	1.23	0	7.59	29	17.2	3.32	0.44	30.8	1.14	1	33.6
5	10.7	4.17	0.13	33.9	1.86	1	1.1	30	5.2	4.50	1.00	32.0	1.54	0	9.25
6	18.7	4.81	0.86	36.7	1.18	1	0.39	31	0.7	2.08	0.25	30.7	1.92	0	373.89
7	17.9	3.85	0.65	47.0	1.49	1	4.59	32	4.9	2.03	0.16	34.8	1.88	0	358.91
8	2.6	2.39	0.83	45.5	1.53	0	146	33	5.8	2.12	0.02	30.2	1.97	0	369.85
9	2.0	4.69	0.67	32.9	1.15	0	5.65	34	8.7	2.02	0.19	33.4	1.89	0	362.51
10	11.1	4.97	0.14	44.2	1.92	1	0.64	35	10.5	2.03	0.35	31.3	1.86	1	358.85
11	18.0	3.99	0.82	40.6	1.98	1	7.19	36	0.4	2.04	0.77	34.9	1.95	0	365.41
12	2.0	2.96	0.24	42.6	1.33	0	6.25	37	4.0	2.05	0.17	32.8	1.85	0	358.41
13	17.3	4.63	0.91	43.3	1.00	1	0.27	38	2.1	2.08	0.45	35.6	1.99	0	359.29
14	11.2	3.71	0.32	33.5	1.72	1	29	39	6.8	2.12	0.07	33.5	1.99	0	358.02
15	8.1	3.75	0.50	35.8	1.69	0	24.6	40	10.3	2.03	0.08	36.1	1.89	1	348.89
16	15.6	2.50	0.23	47.4	1.26	1	77.6	41	2.0	2.04	0.89	33.6	1.84	0	349.3
17	7.5	3.15	0.44	31.5	1.57	0	73.3	42	3.0	2.13	0.33	31.3	1.87	0	347.57
18	8.8	3.45	0.70	44.6	1.81	0	25.9	43	8.1	2.04	0.85	35.0	1.90	0	345.66
19	19.0	2.26	0.04	49.5	1.47	1	159	44	4.3	2.12	1.00	30.4	1.90	0	350.13
20	6.8	4.13	0.50	36.1	1.44	0	11.8	45	7.8	2.00	0.93	30.4	1.82	0	360.47
21	11.9	3.55	0.84	42.0	1.74	1	17.3	46	8.4	2.02	0.55	32.3	1.89	0	365.2
22	6.6	3.58	0.80	37.9	1.06	0	19.4	47	11.8	2.09	0.14	32.0	1.86	1	340.6
23	2.9	2.01	0.76	47.8	1.74	0	306	48	19.8	2.01	0.82	34.2	1.97	1	353.39
24	13.4	2.35	0.10	39.8	1.22	1	139	49	12.9	2.04	0.72	35.0	1.90	1	342.23
25	8.7	2.18	0.54	40.2	1.03	0	147	50	14.0	2.00	0.40	36.3	1.88	1	341.54

its output is used to develop the robust optimization model:

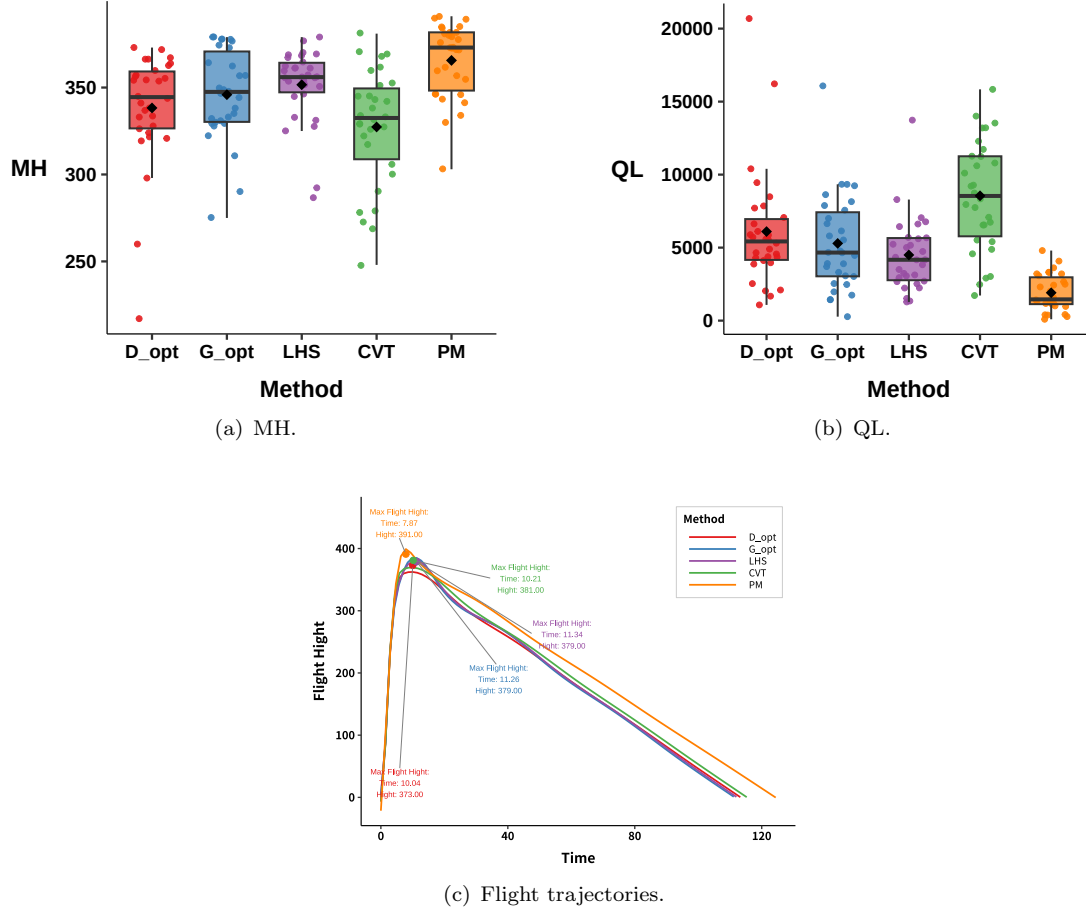
$$L(\mathbf{x}) = (f(\mathbf{x}) - 400)^2 + v(\mathbf{x}), \quad (33)$$

where $f(\mathbf{x})$ is the predicted response, $v(\mathbf{x})$ is the predicted variance, and 400 is the predefined optimal target value. The genetic algorithm is then applied to minimize Equation (33) to obtain the optimal input parameter configuration. The algorithm parameters are set as follows: initial population size of 50, maximum number of iterations of 200, and a stopping threshold of 20, with all other parameters kept at their default values. The optimization process is repeated 30 times, and the actual MH, quality loss (QL) values, and simulated rocket flight trajectories are recorded for each method. The results are presented in Table 6 and Figure 9. In addition, to assess the effect of different initial sample sizes, comparative results for initial training samples of 40, 50, and 60 (corresponding to total training samples of 60, 70, and 80, respectively) are provided in Appendix B.

Figures 9(a-b) clearly illustrates the performance differences among the methods in terms of MH and QL. Compared with the alternatives, the proposed PM method achieves higher mean and median MH values, with fewer outliers. This indicates greater robustness in the optimization process. For QL, the PM method records both a lower

Table 6. MH and QL of different methods.

Method	Mean		Median		Maximum		Minimum		Q1		Q3	
	MH	QL	MH	QL	MH	QL	MH	QL	MH	QL	MH	QL
PM	365.60	1908.09	373.00	1456.03	391.00	4798.06	303.00	91.75	348.25	1128.42	381.75	2967.72
LHS	351.60	4496.26	356.00	4166.58	379.00	13731.95	287.00	1279.43	347.25	2767.04	364.25	5655.17
CVT	327.33	8549.50	332.50	8538.47	381.00	15838.97	248.00	1719.21	308.75	5774.53	349.50	11256.48
D_opt	338.27	6099.77	344.50	5423.04	373.00	20689.30	217.00	1076.50	326.50	4158.41	359.25	6954.43
G_opt	345.90	5288.10	347.50	4660.24	379.00	16077.51	275.00	271.08	330.25	3033.79	370.75	7422.89

**Figure 9.** Optimization results of different methods: (a) MH; (b) QL; (c) Flight trajectories.

mean and median than the other methods. The results from the 30 experiments are also more tightly clustered, with smaller fluctuations, demonstrating superior optimization performance and stability.

As shown in Table 6, the PM method delivers superior performance across all metrics. Compared with the LHS method, its QL mean decreases from 4496.26 to 1908.09 (a reduction of 57.56%), and the median decreases by 65.05%. The maximum and minimum QL values are reduced by 65.06% and 92.83%, respectively. For MH, the mean (365.6) and median (373) increase by 3.98% and 4.78%, with corresponding maximum and minimum improvements of 3.17% and 5.57%. Relative to the CVT method,

the PM method achieves reductions in QL mean and median of 22.30% and 82.95%, respectively, while the MH mean and minimum increase by 11.69% and 22.18%. Compared with the Dopt method, the QL maximum and mean decrease by 76.81% and 68.72%, while the MH mean and median increase by 8.08% and 8.27%. In comparison with the Gopt method, the QL mean, median, and minimum decrease by 63.92%, 68.76%, and 66.16%, respectively, and the MH mean and median increase by 5.69% and 7.34%, with maximum and minimum improvements of 3.17% and 10.18%.

The results demonstrate that the proposed two-stage active learning algorithm with an attention mechanism can more accurately capture the response surface in the potential optimum region and improve the efficiency of design point utilization. This enhancement significantly increases the effectiveness of the response surface model in robust parameter design and strengthens both the robustness and reliability of the optimal solution.

The optimal solution is determined as the design yielding the highest maximum flight height among the 30 optimization runs. Figure 9(c) presents the MH values and corresponding flight trajectories of the optimal solutions obtained by different methods. Compared with the alternatives, the PM method achieves the best performance in terms of MH, reaching 391 m, which is closer to the target value of 400 m. Regarding flight time, all methods reach the peak altitude in approximately 7.5 s and return to the ground in about 122 s. However, the MH values of the other methods are noticeably lower than that of the PM method. Comparison of the trajectory profiles reveals that, although ascent durations are nearly identical across methods, the PM method provides a more robust optimal solution closer to the target value, thereby improving the efficiency of product quality improvement.

4.5. Discussion

This study proposes a two-stage active learning algorithm based on an attention mechanism. Its core lies in constructing an attention model that can dynamically focus on both the potential optimum region and the space-filling performance of design points. This mechanism guides the allocation of newly added design points to the target subregion in a prioritized and balanced manner. As a result, the predictive accuracy within the potential optimum region is substantially improved, thereby enhancing both the robustness and the effectiveness of the optimal solution.

Although the method strengthens local modeling at the cost of a slight reduction in global predictive accuracy, this does not undermine its overall effectiveness. Prioritizing new design points in the potential optimum region markedly improves accuracy in that region and provides a modest positive effect on overall prediction. Because accuracy in the potential optimum region largely determines optimization quality, focusing computational and experimental resources there is essential for effective robust optimization.

Due to the attention-guided iterative updates, computational cost is slightly higher than classical designs (e.g., LHS) but comparable to other active learning methods. The method achieves higher modeling and optimization efficiency for robust parameter design and provides a practical solution for nonstationary response-surface modeling under small-sample conditions.

5. Conclusion

To address quality improvement with nonstationary responses under small-sample conditions, an attention-based active learning NSGP modeling and robust optimization framework is proposed. First, a Bayesian hierarchical specification is used to construct the NSGP structure, capturing input-dependent nonstationarity between design variables and output responses. Next, a two-stage attention mechanism is designed to balance quality loss with space-filling, guiding new design points toward the potential optimum region. Finally, the training set and NSGP model are updated, and a quality-loss-based robust optimization model is formulated to obtain the optimal input parameter settings. The key advantage is the two-stage attention-weighted sampling strategy, which efficiently detects the potential optimum region of a nonstationary response, allocates experimental resources to that region, and enhances predictive accuracy in critical areas for optimization. Consequently, design points are used more efficiently and the nonstationary response surface model is more effective. This, in turn, leads to more robust optimization results.

It should be noted that the proposed method adopts a focused attention strategy that prioritizes predictive accuracy in the potential optimum region to improve efficiency in robust parameter design. Global predictive accuracy is not an explicit objective, which may limit applicability to other optimization tasks. Future research is expected to explore multi-level attention mechanisms that jointly consider global and local accuracy, thereby extending the applicability of the proposed approach.

Disclosure statement

The authors declare no competing interests.

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Data availability statement

The authors confirm that the data supporting the findings of this study are available within the article.

References

- Alshraideh, H., and E. del Castillo. 2014. "Gaussian process modeling and optimization of profile response experiments." *Quality and Reliability Engineering International* 30 (4): 449–462.
- An, Y., D. Wang, and X. Zhang. 2020. "A novel local temperature change detection approach in a 3D thermal field." *Quality Technology & Quantitative Management* 17 (6): 723–737.
- Ankenman, B., B. L. Nelson, and J. Staum. 2010. "Stochastic Kriging for simulation meta-modeling." *Operations Research* 58 (2): 371–382.
- Bao, L. L., Q. Huang, and K. B. Wang. 2016. "Robust parameter design for profile quality control." *Quality and Reliability Engineering International* 32 (3): 1059–1070.
- Barde, S.. 2024. "Bayesian estimation of large-scale simulation models with Gaussian process regression surrogates." *Computational Statistics & Data Analysis* 196: 107972.
- Box, S., C. M. Bishop, and H. Hunt. 2011. "Stochastic six-degree-of-freedom flight simulator for passively controlled high-power rockets." *Journal of Aerospace Engineering* 24 (1): 31–45.

- Chen, G., and R. Tuo. 2022. "Projection pursuit Gaussian process regression." *IIE Transactions* 55 (9): 901–911.
- Chen, J. H., L. L. Kang, and G. Lin. 2021. "Gaussian process assisted active learning of physical laws." *Technometrics* 63 (3): 329–342.
- Chuang, C. J., and C. W. Wu. 2017. "Determining optimal process mean and quality improvement in a profit-maximization supply chain model." *Quality Technology & Quantitative Management* 16 (2): 154–169.
- Davis, C. B., C. M. Hans, and T. J. Santner. 2021. "Prediction of non-stationary response functions using a Bayesian composite Gaussian process." *Computational Statistics & Data Analysis* 154: 107083.
- Dellino, G., J. P. C. Kleijnen, and C. Meloni. 2012. "Robust optimization in simulation: Taguchi and Krige combined." *INFORMS Journal on Computing* 24 (3): 471–484.
- Ding, N., Z. He, S. He, and L. Song. 2024. "Real-time profile monitoring schemes considering covariates using Gaussian process via sensor data." *Quality Technology & Quantitative Management* 21 (1): 35–53.
- Ding, R., D. K. J. Lin, and D. Wei. 2004. "Dual-response surface optimization: a weighted MSE approach." *Quality Engineering* 16 (3): 377–385.
- Fan, S. K. S. 2000. "A generalized global optimization algorithm for dual response systems." *Journal of Quality Technology* 32 (4): 444–456.
- Feng, Z., J. Wang, Y. Ma, and Y. Tu. 2021. "Robust parameter design based on Gaussian process with model uncertainty." *International Journal of Production Research* 59 (9): 2772–2788.
- Feng, Z., J. Wang and W. Huang. 2025. "An active learning multi-objective robust optimization framework via Gaussian process models under uncertainty." *Engineering Optimization*: 1–24.
- Gelman, A., H. S. Stern, J. B. Carlin, D. B. Dunson, A. Vehtari, and D. B. Rubin. 2013. *Bayesian Data Analysis*. Chapman and Hall/CRC.
- Gnanasambandam, R., B. Shen, A. C. C. Law, C. Dou, and Z. Kong. 2024. "Deep Gaussian process for enhanced Bayesian optimization and its application in additive manufacturing." *IIE Transactions*: 1–14.
- Gramacy, R. B., and H. K. H. Lee. 2008. "Bayesian treed Gaussian process models with an application to computer modeling." *Journal of the American Statistical Association* 103 (483): 1119–1130.
- Guo, B., X. Li, M. Liu, and X. Yang. 2023. "Construction of orthogonal general sliced Latin hypercube designs." *Statistical Papers* 64 (3): 987–1014.
- Han, M., Q. Huang, L. Ouyang, and X. Zhao. 2023. "A kriging-based active learning algorithm for contour estimation of integrated response with noise factors." *Engineering with Computers* 39(2): 1341–1362.
- Han, M., and M. H. Y. Tan. 2016. "Integrated parameter and tolerance design with computer experiments." *IIE Transactions* 48 (11): 1004–1015.
- He, Z., Y. Gao, L. Qu, and Z. Wang. 2021. "A nonparametric CUSUM scheme for monitoring multivariate time-between-events-and-amplitude data with application to automobile painting." *International Journal of Production Research* 60 (18): 5432–5449.
- Heinonen, M., H. Mannerström, J. Rousu, S. Kaski, and H. Lähdesmäki. 2016. "Non-stationary Gaussian process regression with Hamiltonian Monte Carlo." *PMLR*: 732–740.

- Hoffman, M. D., and A. Gelman. 2014. "The No-U-Turn sampler: adaptively setting path lengths in Hamiltonian Monte Carlo." *Journal of Machine Learning Research* 15 (1): 1593–1623.
- Huang, Q.. 2022. "An impulse response formulation for small-sample learning and control of additive manufacturing quality." *IISE Transactions* 55 (9): 926–939.
- Jiang, F., and M. H.-Y. Tan. 2021. "Shifted log loss Gaussian process model for expected quality loss prediction in robust parameter design." *Quality Technology & Quantitative Management* 18 (5): 527–551.
- Kleijnen, J. P. C., and E. Mehdad. 2014. "Multivariate versus univariate Kriging metamodels for multi-response simulation models." *European Journal of Operational Research* 236 (2): 573–582.
- Konomi, B., G. Karagiannis, A. Sarkar, X. Sun, and G. Lin. 2014. "Bayesian treed multivariate Gaussian process with adaptive design: Application to a carbon capture unit." *Technometrics* 56 (2): 145–158.
- Li, A., Z. He, Q. Wang, Y. Zhang, and Y. Ma. 2025a. "A multi-objective evolutionary algorithm with mutual-information-guided improvement phase for feature selection in complex manufacturing processes." *European Journal of Operational Research* 323 (3): 952–965.
- Li, T., and J. Ma. 2022. "Attention mechanism based mixture of Gaussian processes." *Pattern Recognition Letters* 161: 130–136.
- Li, W., J. Qin, C. Wu and F. Tsung. 2025b. "Online monitoring of directed count-weighted network with attributes via the gravity model." *Quality Technology & Quantitative Management*: 1-16.
- Li, Y., and Q. Zhou. 2016. "Pairwise meta-modeling of multivariate output computer models using nonseparable covariance function." *Technometrics* 58 (4): 483–494.
- Lichota, P.. 2016. "Inclusion of the D-optimality in multisine manoeuvre design for aircraft parameter estimation." *Journal of Theoretical and Applied Mechanics* 54 (1): 87–98.
- Liu, Y., D. Huang, B. Liu, Q. Feng, and B. Cai. 2021. "Adaptive ranking based ensemble learning of Gaussian process regression models for quality-related variable prediction in process industries." *Applied Soft Computing* 101: 107060.
- Liu, Z., Y. Li, X. Yue, and E. Pan. 2025. "Latent functional Gaussian process incorporating output spatial correlations." *IISE Transactions*: 1–14.
- Meng, Q., S. Wang, and S. H. Ng. 2022. "Combined global and local search for optimization with Gaussian process models." *INFORMS Journal on Computing* 34 (1): 622–637.
- Mustafa, H., C. Bourelly, M. Vitelli, F. Milano, M. Molinara, and L. Ferrigno. 2024. "SoC estimation on Li-ion batteries: A new EIS-based dataset for data-driven applications." *Data Brief* 57: 110947.
- Omwando, C. N.. 2024. "G-criterion for second order rotatable designs constructed using trigonometric transformations." *American Journal of Applied Mathematics and Statistics* 12 (3): 70–74.
- Ouyang, L., Y. Ling, L. Liu, M. Sun, and M. Wang. 2025. "Double-robust Bayesian variable selection and model prediction with spherically symmetric error." *IISE Transactions*: 1–26.
- Ouyang, L., Y. Ma, J. H. Byun, J. Wang, and Y. Tu. 2016. "An interval approach to robust design with parameter uncertainty." *International Journal of Production Research* 54 (11): 3201–3215.

- Peterson, J. J., G. Miró-Quesada, and E. del Castillo. 2009. “A Bayesian reliability approach to multiple response optimization with seemingly unrelated regression models.” *Quality Technology & Quantitative Management* 6 (4): 353–369.
- Rhode, S.. 2020. “Non-stationary Gaussian process regression applied in validation of vehicle dynamics models.” *Engineering Applications of Artificial Intelligence* 93: 103716.
- Santner, T. J., B. J. Williams, and W. I. Notz. 2018. *Space-filling designs for computer experiments*. New York: Springer.
- Sauer, A., R. B. Gramacy, and D. Higdon. 2022. “Active learning for deep Gaussian process surrogates.” *Technometrics* 65 (1): 4–18.
- Tan, M. H. Y., and C. F. J. Wu. 2012. “Robust design optimization with quadratic loss derived from Gaussian process models.” *Technometrics* 54 (1): 51–63.
- Vassiliades, V., K. Chatzilygeroudis, and J. Mouret. 2018. “Using centroidal voronoi tessellations to scale up the multidimensional archive of phenotypic elites algorithm.” *IEEE Transactions on Evolutionary Computation* 22 (4): 623–630.
- Wang, C., J. Xiao, X. Huang, and J. Xu. 2025. “RSDS: an efficient adaptive Kriging-based method for structural reliability analysis considering the distance between sample points.” *Quality Technology & Quantitative Management*: 1–27.
- Wang, H., J. Yuan, and S. H. Ng. 2019. “Gaussian process based optimization algorithms with input uncertainty.” *IIE Transactions* 52 (4): 377–393.
- Wang, W., X. Yue, B. Haaland, and C. F. J. Wu. 2022. “Gaussian processes with input location error and applications to the composite parts assembly process.” *SIAM/ASA Journal on Uncertainty Quantification* 10 (2): 619–650.
- Williams, C. K. I., and C. E. Rasmussen. 2006. *Gaussian Processes for Machine Learning*. Cambridge: MIT Press.
- Wu, C. F. J., and M. S. Hamada. 2011. *Experiments: Planning, Analysis and Parameter Design Optimization*. New York: John Wiley & Sons.
- Wu, C. W., M. H. Shu, and N. Y. Wu. 2020. “Acceptance sampling schemes for two-parameter Lindley lifetime products under a truncated life test.” *Quality Technology & Quantitative Management* 18 (3): 382–395.
- Xie, J., Z. Ma, D. Chang, G. Zhang, and J. Guo. 2022. “GPCA: A probabilistic framework for Gaussian process embedded channel attention.” *IEEE Transactions on Pattern Analysis and Machine Intelligence* 44 (11): 8230–8248.
- Yu, J., S. Li, X. Liu, Y. Gao, S. Wang, and C. Liu. 2022. “Dynamic convolutional gated recurrent unit attention auto-encoder for feature learning and fault detection in dynamic industrial processes.” *International Journal of Production Research* 61 (21): 7434–7452.
- Zhang, W., Z. Zhao, H. Xu, X. Li, and Z. Wang. 2024. “AK-Gibbs: An active learning Kriging model based on Gibbs importance sampling algorithm for small failure probabilities.” *Computer Methods in Applied Mechanics and Engineering* 426: 116992.
- Zhong, Y., Z. Ling, L. Liu, S. Zhang, and H. Wen. 2024. “Deep Gaussian attention network for lumber surface defect segmentation.” *IEEE Transactions on Instrumentation and Measurement* 73: 1–12.
- Zhou, M., R. Zuo, C. Sung, Y. Tong, and X. Wang. 2025. “Region-optimal Gaussian process surrogate model via Dirichlet process for cold-flow and combustion emulations.” *Computer Methods in Applied Mechanics and Engineering* 439: 117894.

6. Appendices

Appendix A.

Table A1. Maximum RMSE for 10 active learning samples.

Number	D_opt	G_opt	LHS	CVT	PM
40	0.1976	0.1426	0.2912	0.1105	0.1619
50	0.1206	0.0990	0.0935	0.0777	0.0840
60	0.0871	0.0824	0.0679	0.1891	0.0544
70	0.0826	0.0801	0.0815	0.0784	0.0490
80	0.0581	0.1178	0.0539	0.0565	0.0440
90	0.0566	0.0465	0.0409	0.0618	0.0480
100	0.0313	0.0371	0.0365	0.0510	0.0290

As shown in Table A1, the PM method consistently achieves superior control over the maximum RMSE, and its advantage becomes increasingly evident as the sample size grows. For example, at 40 total samples, the maximum RMSE is 0.1619, representing a 44.40% reduction compared with LHS (0.2912). When the sample size increases to 100, the maximum value further decreases to 0.0290—the lowest among all methods—corresponding to a 7.35% improvement over D_opt (0.0313) and a 43.14% improvement over CVT (0.0510). The largest improvement is observed at 60 samples, where PM outperforms CVT by 71.23%.

Table A2. Minimum RMSE for 10 active learning samples.

Number	D_opt	G_opt	LHS	CVT	PM
40	0.0398	0.0639	0.0265	0.0381	0.0213
50	0.0143	0.0232	0.0260	0.0167	0.0126
60	0.0302	0.0064	0.0051	0.0332	0.0210
70	0.0012	0.0025	0.0234	0.0029	0.0047
80	0.0191	0.0008	0.0212	0.0208	0.0134
90	0.0017	0.0204	0.0048	0.0167	0.0122
100	0.0003	0.0167	0.0067	0.0016	0.0004

As shown in Table A2, the PM method also demonstrates strong competitiveness in terms of minimum RMSE. With 40 samples, PM attains the lowest value of 0.0213, improving by 19.62% compared with LHS (0.0265) and by 66.67% compared with G_opt (0.0639). At 100 samples, the minimum RMSE decreases to 0.0004—comparable to D_opt (0.0003) but 97.60% better than G_opt (0.0167). Although some competing methods slightly outperform PM in certain sample sizes (e.g., 70 and 90), PM maintains a leading position in most cases.

Table A3. Q1 of RMSE for 10 active learning samples.

Number	D_opt	G_opt	LHS	CVT	PM
40	0.0665	0.0729	0.0587	0.0490	0.0381
50	0.0468	0.0542	0.0488	0.0522	0.0322
60	0.0425	0.0481	0.0371	0.0390	0.0319
70	0.0380	0.0372	0.0308	0.0298	0.0274
80	0.0308	0.0315	0.0281	0.0263	0.0212
90	0.0253	0.0275	0.0231	0.0231	0.0177
100	0.0210	0.0217	0.0188	0.0211	0.0153

Table A4. Q3 of RMSE for 10 active learning samples.

Number	D_opt	G_opt	LHS	CVT	PM
40	0.1066	0.1070	0.0908	0.0794	0.0597
50	0.0741	0.0721	0.0686	0.0661	0.0441
60	0.0578	0.0604	0.0538	0.0585	0.0420
70	0.0483	0.0523	0.0442	0.0408	0.0358
80	0.0415	0.0452	0.0363	0.0394	0.0285
90	0.0369	0.0357	0.0314	0.0302	0.0238
100	0.0270	0.0274	0.0268	0.0283	0.0222

To evaluate robustness and consistency, the interquartile range ($\text{IQR} = \text{Q3} - \text{Q1}$) derived from Tables A3 and A4 is further analyzed. A smaller IQR indicates a more concentrated and reliable distribution. At 40 samples, PM achieves the smallest IQR of 0.0216 ($0.0597 - 0.0381$), improving concentration by 28.95% over CVT (0.0304) and by 46.13% over D_opt (0.0401). At 100 samples, the IQR further decreases to 0.0069 ($0.0222 - 0.0153$), maintaining high robustness. The largest robustness advantage occurs at 50 samples, where the IQR is 0.0119, representing a 56.41% improvement over D_opt (0.0273).

Table A5. Maximum RMSE for 20 active learning samples.

Number	D_opt	G_opt	LHS	CVT	PM
40	0.3059	0.2248	0.1732	0.1170	0.2559
50	0.1864	0.0895	0.1043	0.1075	0.2452
60	0.1259	0.0999	0.0664	0.0899	0.0404
70	0.0758	0.0623	0.0574	0.0707	0.0442
80	0.0750	0.0621	0.0617	0.0670	0.0388
90	0.0538	0.0481	0.0468	0.0609	0.0512
100	0.0416	0.0485	0.0462	0.0601	0.0390

As shown in Table A5, the PM method consistently yields the lowest maximum

RMSE across all sample sizes. The most significant improvement occurs at 60 samples, where PM reduces the maximum RMSE by 67.91% compared with D_opt. Even at 100 samples, PM still achieves a 6.25% reduction.

Table A6. Minimum RMSE for 20 active learning samples.

Number	D_opt	G_opt	LHS	CVT	PM
40	0.0614	0.0402	0.0287	0.0559	0.0086
50	0.0336	0.0058	0.0063	0.0364	0.0009
60	0.0048	0.0232	0.0254	0.0039	0.0081
70	0.0067	0.0257	0.0129	0.0309	0.0071
80	0.0003	0.0187	0.0187	0.0044	0.0048
90	0.0147	0.0144	0.0180	0.0189	0.0002
100	0.0147	0.0145	0.0073	0.0205	0.0008

Table A6 shows that when the total sample size is 40, PM achieves a minimum RMSE of 0.0086, outperforming all alternatives. The best overall performance is obtained at 50 samples, with a minimum RMSE of 0.0009. The greatest improvement is observed at 90 samples, where PM reduces the minimum RMSE by 98.64% compared with D_opt. The smallest gain, at 40 samples, still reaches 69.00% relative to LHS.

Table A7. Q1 of RMSE for 20 active learning samples.

Number	D_opt	G_opt	LHS	CVT	PM
40	0.1081	0.0765	0.0482	0.0741	0.0203
50	0.0701	0.0449	0.0401	0.0635	0.0185
60	0.0337	0.0404	0.0291	0.0505	0.0162
70	0.0341	0.0326	0.0269	0.0431	0.0166
80	0.0261	0.0308	0.0305	0.0369	0.0154
90	0.0240	0.0228	0.0214	0.0340	0.0135
100	0.0232	0.0285	0.0182	0.0266	0.0121

Table A8. Q3 of RMSE for 20 active learning samples.

Number	D_opt	G_opt	LHS	CVT	PM
40	0.1775	0.1003	0.0861	0.0883	0.0584
50	0.1237	0.0599	0.0665	0.0782	0.0297
60	0.0611	0.0553	0.0457	0.0639	0.0242
70	0.0538	0.0463	0.0438	0.0600	0.0270
80	0.0418	0.0514	0.0413	0.0482	0.0302
90	0.0395	0.0346	0.0299	0.0443	0.0230
100	0.0323	0.0358	0.0271	0.0388	0.0193

Tables A7 and A8 further confirm that PM maintains the smallest interquartile range ($IQR = Q3 - Q1$) for all sample sizes. This indicates better robustness and concentration. The most pronounced improvement appears at 50 samples, with a 79.10% gain over D_opt. Even at 100 samples, PM achieves a 1.37% improvement compared with G_opt. These results demonstrate that PM maintains high stability and reliability under varying sample conditions.

Appendix B.

Table B1. Comparison of optimization results with 60 training samples.

Method	Mean		Median		Maximum		Minimum		Q1		Q3	
	MH	QL	MH	QL	MH	QL	MH	QL	MH	QL	MH	QL
PM	365.83	2930.53	371.50	3013.73	392.00	6720.97	300.00	1018.22	350.00	2160.40	383.00	3456.65
LHS	351.87	5784.58	353.50	5342.49	377.00	9498.79	320.00	2690.94	347.00	4459.98	359.00	7648.59
CVT	325.97	8622.45	339.50	7396.28	365.00	21212.57	246.00	777.46	307.00	6151.12	353.50	10521.37
D_opt	328.20	7570.85	340.50	6076.41	388.00	19800.33	239.00	1474.00	301.25	4308.73	357.50	9276.37
G_opt	334.27	7235.88	338.00	6161.14	391.00	17281.67	251.00	1879.86	319.00	4146.96	356.25	9606.40

As shown in Table B1, the PM method consistently outperforms other approaches with 60 training samples. Compared with LHS, the PM method reduces the QL mean from 5784.58 to 2930.53, achieving a 49.34% decrease. The median and minimum QL values are also reduced by 43.59% and 62.16%, respectively. For the MH metric, the PM method improves the mean, median, and maximum by 3.97%, 5.09%, and 3.98%. Relative to CVT, PM shows even stronger performance. It reduces the QL mean by 66.01%, the median by 59.25%, and the maximum by 68.32%. Meanwhile, MH performance improves by 12.23% in the mean and 21.95% in the minimum. When compared with D_opt, PM lowers the QL mean by 61.29% and the maximum by 66.06%. It also increases the MH mean by 11.47% and the median by 9.10%. Compared with G_opt, the QL mean, median, maximum, and minimum are reduced by 59.50%, 51.08%, 61.10%, and 45.84%, respectively. Additionally, the MH mean and median are improved by 9.44% and 9.91%, while the minimum increases by 19.52%.

Table B2. Comparison of optimization results with 70 training samples.

Method	Mean		Median		Maximum		Minimum		Q1		Q3	
	MH	QL	MH	QL	MH	QL	MH	QL	MH	QL	MH	QL
PM	369.53	3527.62	374.50	2863.82	392.00	8610.52	318.00	1854.14	360.50	2516.13	380.50	4044.50
LHS	346.00	7056.32	352.00	6489.23	384.00	19136.57	242.00	1914.78	334.50	4823.50	362.25	8758.49
CVT	336.07	8476.93	341.50	7220.50	385.00	26473.52	261.00	2497.03	327.50	5729.68	351.00	9735.58
D_opt	335.33	8640.93	338.50	7532.22	374.00	25676.47	233.00	3308.62	323.75	4919.09	356.25	10550.84
G_opt	334.07	7835.69	335.00	5989.16	387.00	18779.07	275.00	2748.34	313.75	4785.09	361.00	10509.83

As shown in Table B2, with 70 training samples, the PM method consistently outperforms all other methods across key indicators. Compared with LHS, PM reduces the QL mean from 7056.32 to 3527.62, achieving a 50.00% reduction. The median and

maximum QL values are reduced by 55.88% and 55.00%, respectively. Meanwhile, MH improves by 6.80% in the mean, 6.39% in the median, and 31.40% in the minimum. Against CVT, PM lowers the QL mean by 58.38% and the maximum by 67.48%. The MH mean increases by 9.96%, while the minimum rises by 21.84%. Compared with D_opt, PM achieves a 59.18% reduction in QL mean and a 66.46% drop in the maximum. At the same time, MH improves by 10.20% in the mean and 36.48% in the minimum. Relative to G_opt, QL values are significantly lower. The mean and median are reduced by 54.98% and 52.18%, respectively. In terms of MH, the mean increases by 10.61%, the median by 11.79%, and the minimum by 15.64%.

Table B3. Comparison of optimization results with 80 training samples.

Method	Mean		Median		Maximum		Minimum		Q1		Q3	
	MH	QL	MH	QL	MH	QL	MH	QL	MH	QL	MH	QL
PM	366.60	5597.53	372.50	4794.50	392.00	14924.09	319.00	2131.16	356.00	3481.38	378.00	5943.41
LHS	350.67	8907.73	356.00	8332.60	384.00	21441.47	272.00	3271.37	343.00	6424.00	361.00	10717.41
CVT	335.57	10070.66	336.50	9343.45	371.00	18541.11	277.00	2641.21	323.50	7482.70	352.00	12569.66
D_opt	319.40	12093.48	323.00	10690.87	372.00	22646.91	252.00	3865.38	305.00	8724.00	349.75	16243.79
G_opt	334.57	9588.88	342.00	8141.33	389.00	21515.31	235.00	3256.61	322.75	6710.74	359.75	12136.71

As shown in Table B3, with 80 training samples, the PM method continues to outperform all competing methods. Compared with LHS, PM reduces the QL mean from 8907.73 to 5597.53, representing a 37.16% reduction. The median QL is also reduced by 42.46%. In terms of MH, the mean and median increase by 4.54% and 4.63%, respectively, while the minimum rises by 17.28%. Relative to CVT, PM lowers the QL mean by 44.42% and the median by 48.69%. It also improves the MH mean by 9.25% and the minimum by 15.16%. Compared with D_opt, PM achieves a 53.72% reduction in QL mean. The MH mean and median increase by 14.78% and 15.32%, respectively, and the minimum improves by 26.59%. Against G_opt, PM achieves a 41.62% reduction in QL mean and a 41.11% decrease in median. MH performance also improves, with the mean and median increasing by 9.57% and 8.92%, respectively, and the minimum rising by 35.74%.